Real-Time Optimization and Maintenance of Wind Turbine Performance Using Digital Twin Technology

Meet Our Team !

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INTRODUCTION

Real-Time Optimization and Maintenance of Wind Turbine Performance Using Digital Twin Technology

- Wind energy is one of the most rapidly growing renewable energy sources
- reducing greenhouse gas emissions, providing a clean, sustainable source of power.
- Mainitaning turbines efficiently is a challenge due to their size, complexity, and remote locations.
- A **Digital Twin** is a virtual replica of a physical asset, system, or process.

RESEARCH QUESTIONS

- **Weather Impact Analysis**
  How can weather conditions such as wind speed, wind direction, and lightning events be modeled and analyzed through a digital twin to predict their impact on wind turbine performance and power generation?

- **Noise Effect Investigation**
  How can a digital twin with machine learning optimize wind turbine noise and performance in real time?

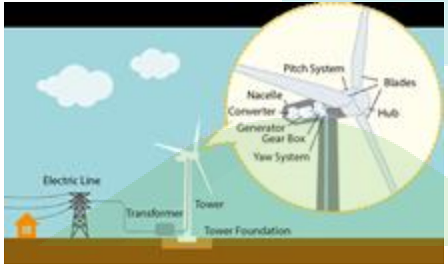
- **Maintenance Optimization**
  How Predict and plan maintenance efficiently?

- **Power Efficiency Improvement**
  In what ways can a Digital Twin system improve powermonitoring, optimize energy output, and enhanceforecasting capabilities for wind turbines?

OBJECTIVES

Main Objective

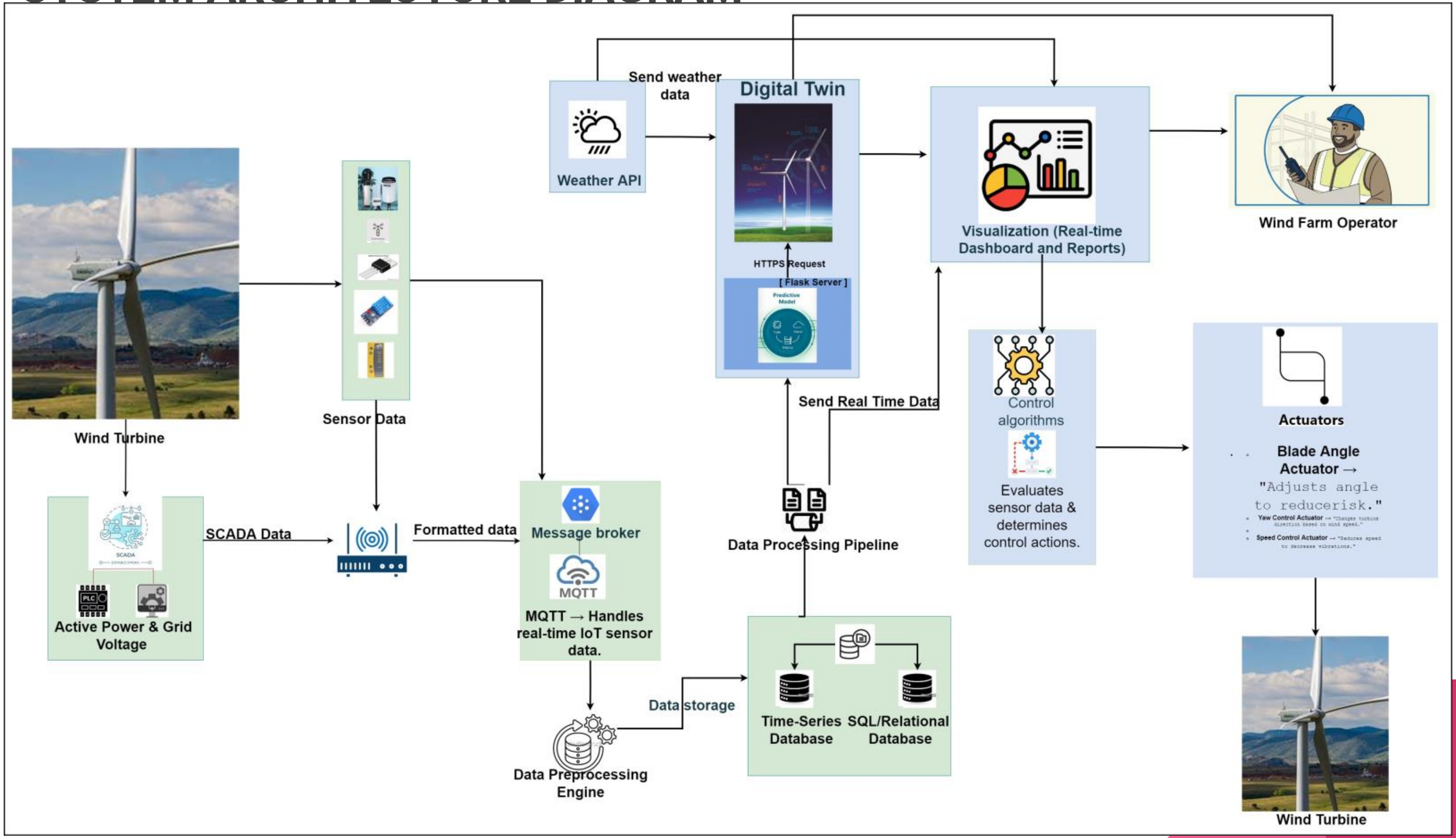
Real-Time Optimization and Maintenance of Wind Turbine Performance Using Digital Twin Technology.



Sub Objectives

1. Analyzing weather impacts on turbines.
2. Enhancing power generation efficiency.
3. Investigating noise effects in wind turbines.
4. Optimizing wind turbine maintenance strategies.

SYSTEM ARCHITECTURE DIAGRAM



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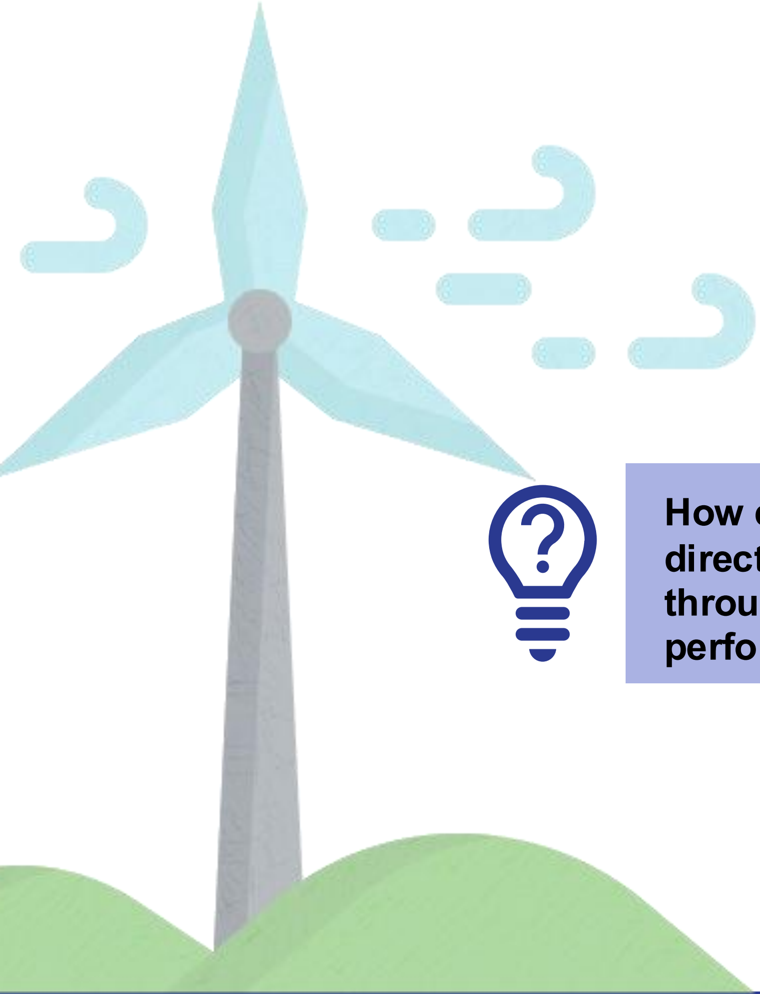
Specialization – Software Engineering

Weather Impact Analysis

SLIIT
FACULTY OF COMPUTING



INTRODUCTION

Research Question

How can weather conditions such as wind speed, wind direction, and lightning events be modeled and analyzed through a digital twin to predict their impact on wind turbine performance and power generation?

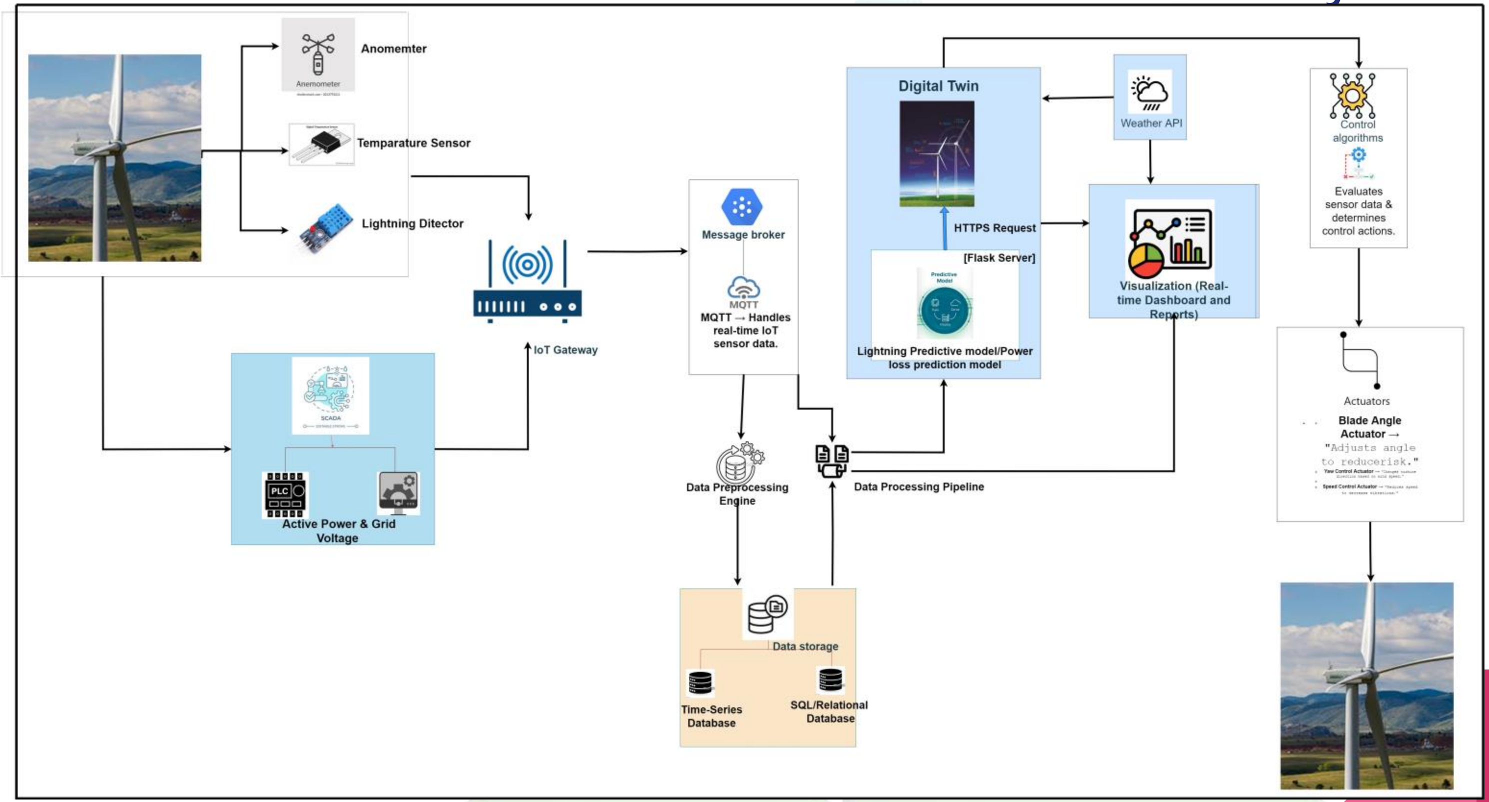
INTRODUCTION

Research Question

- **Analyzing Weather Impact on Wind Turbines Using Digital Twin Technology**

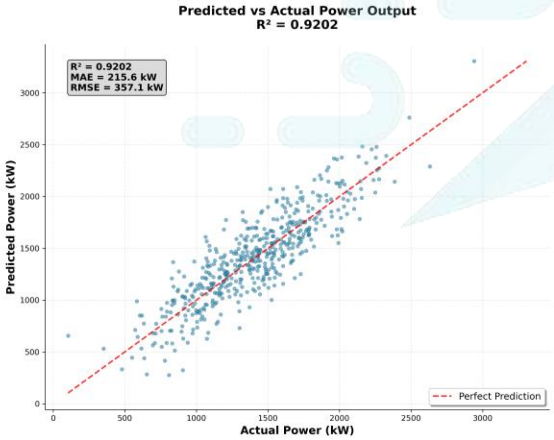
1. Analyze weather and SCADA datasets to identify key characteristics and patterns.
2. Clean, Preprocess data and select suitable ML algorithms, for prediction tasks.
3. Train and evaluate ML models for wind direction, wind speed, forecast power and lightning risk.
4. Build a real-time dashboard to visualize turbine performance and weather impact.
5. Link predictive models and SCADA data using a Flask backend.

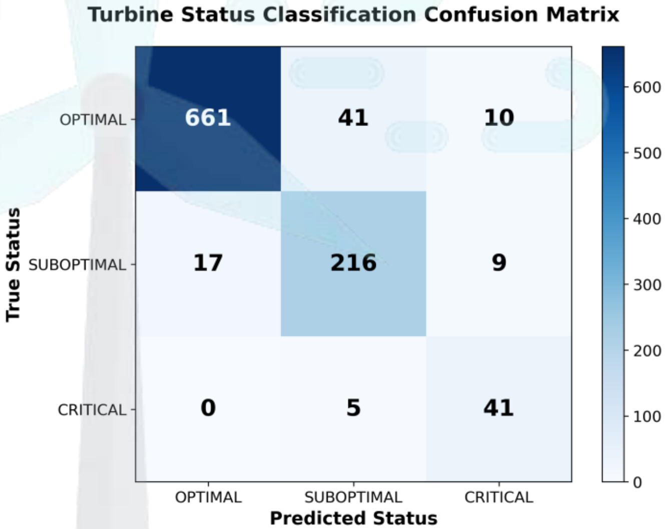
Methodology
System Diagram



Evidence of Completion

1.Turbine Performance Monitoring using Power Prediction



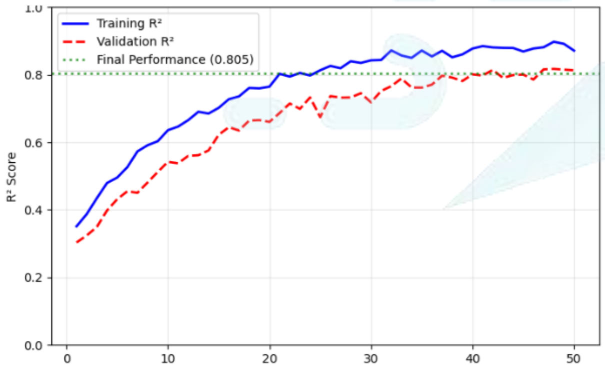


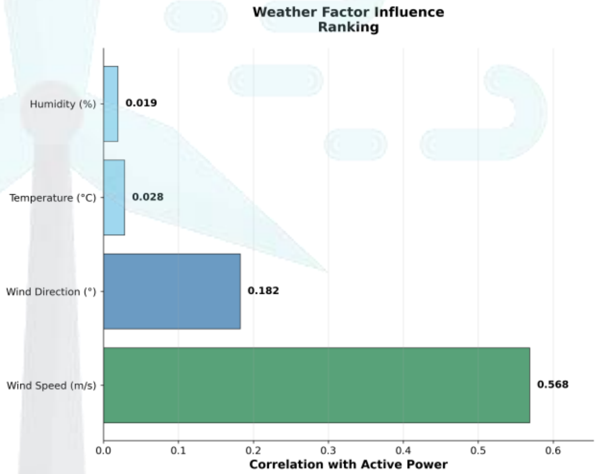
- Best ML Algorithm – **XG Boost Regressor**
- Training data - 47,387(80%)
- Testing data - 5,924(20%)
- No of features - 19

Model	Test R^2	Gap	MAE	RMSE
XGBoost	0.9202	0.0231	215.6	357.1
Random Forest	0.8765	0.0211	245.3	398.2
Gradient Boosting	0.9121	0.0208	225.8	375.4

Evidence of Completion

2. Predicting Turbine Performance under Given Weather Conditions



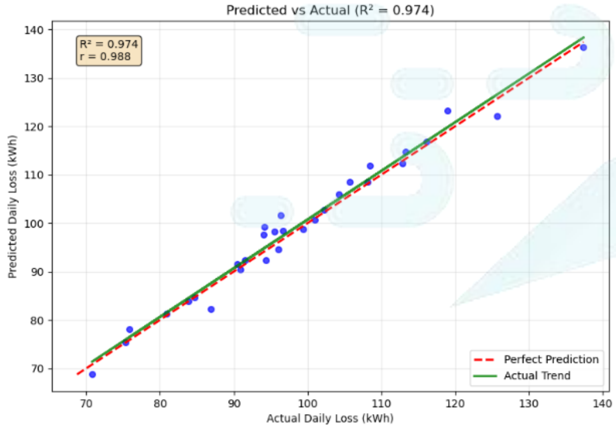


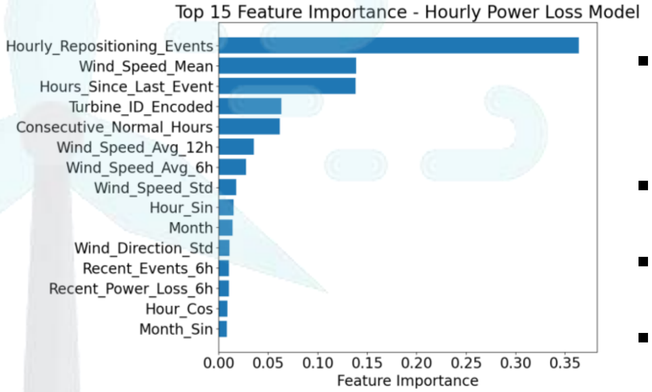
Metric	Value
Overall R^2 Score	0.805
Performance Level	Good (70-85%)
Model Type	Random Forest Multi-Output
Training Records	524,200
Prediction Accuracy	80.5%

- Best ML Algorithm **– Random Forest (MultiOutputRegressor)**
- Training data - 42,160(80%)
- Testing data – 10,539(20%)
- No of features - 8

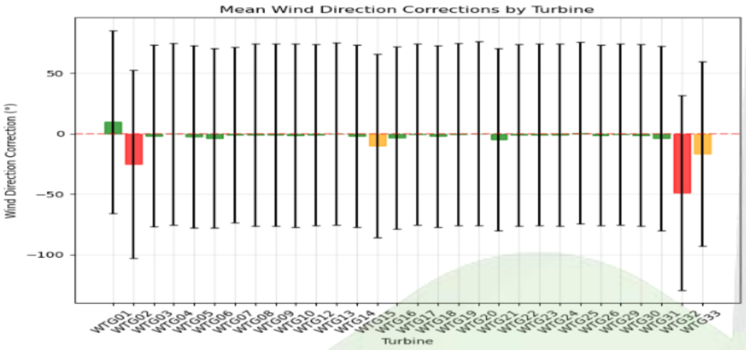
Evidence of Completion

3. Power Loss Due to Wind Direction Change



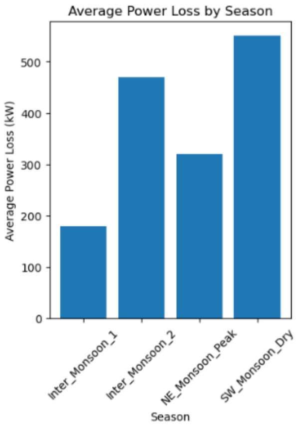


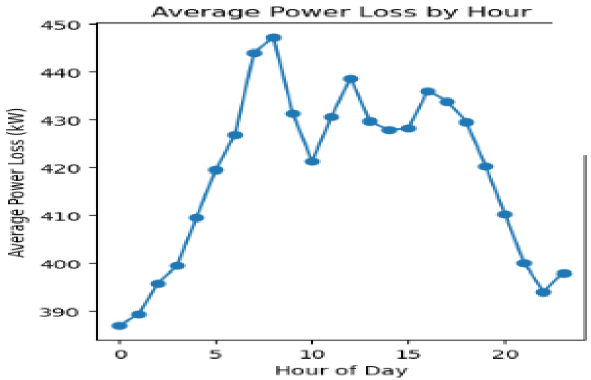
- Best ML Algorithm – **XGBoost Regressor**
- Training data - 42,160(80%)
- Testing data – 10,539(20%)
- No of features – 15
- Average hourly loss: 16.235 kWh

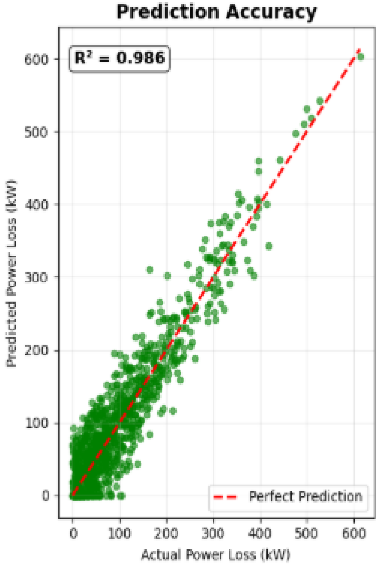


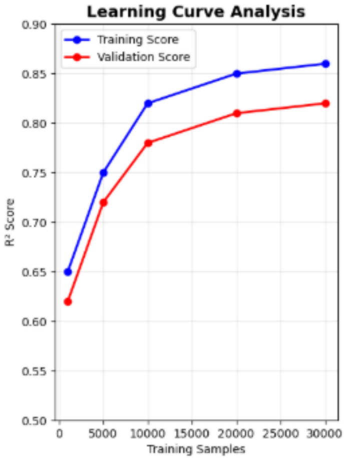
Evidence of Completion

4. Detect Extreme Wind Speeds and Predict Power Loss









Model Summary

Metric	Value
Model Type	XGBoost
Training Samples	524,260
Features	14
R^2 Score	0.986
MAE	26.6 kW
RMSE	75.0 kW
CV Score	0.949 ± 0.028
Turbines	10

Evidence of Completion

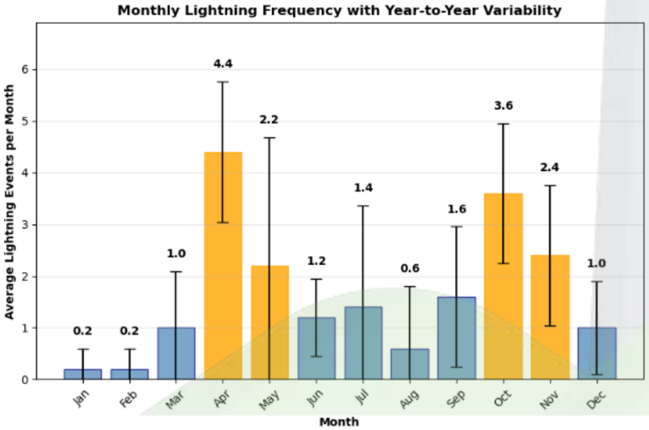
5. Lightning Risk Assessment

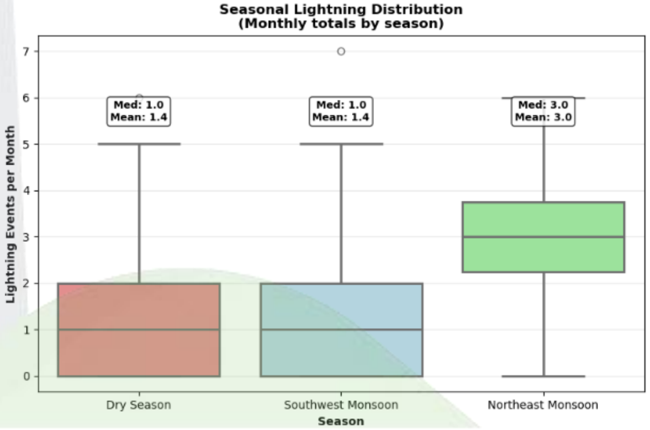
```
ANNUAL TRENDS
   Annual Trend: +0.40 events/year
   2019: 21 events
   2020: 18 events
   2021: 16 events
   2022: 24 events
   2023: 20 events

LIGHTNING EVENT CHARACTERISTICS
   Average Flash Count: 10.03
   Average Radiance: 650985.64
   Average Distance: 67.07 km
   Peak Lightning Hour: 11:00 (12 events)

OVERALL STATISTICS (2019-2023)
   Total Lightning Events: 99
   Total Hours Analyzed: 43,824
   Lightning Frequency: 0.2259% of hours
   Average Events per Day: 0.05
```

Metric	Value
Model Type	Random Forest
Training Samples	5,843 (6-hour windows)
Test Samples	1,460 (2023 test set)
Lightning Events	99 total events
AUC Score	0.999 (Excellent)
Precision	3.7% (Optimized)
Recall	85% (Safety-focused)
F1 Score	0.071
Detection Rate	17/20 events caught
Warning Frequency	31.2% of periods





Methodology
Evidence of Completion

- Data Collection
- Data Preprocessing
- Model Selection
- Model Training & Hyperparameter Tuning
- Model Evaluation
- Display Results Using Web UI

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Specialization - Software engineering

Introduction - Background

In what ways can a Digital Twin system improve power monitoring, optimize energy output, and enhance forecasting capabilities for wind turbines?

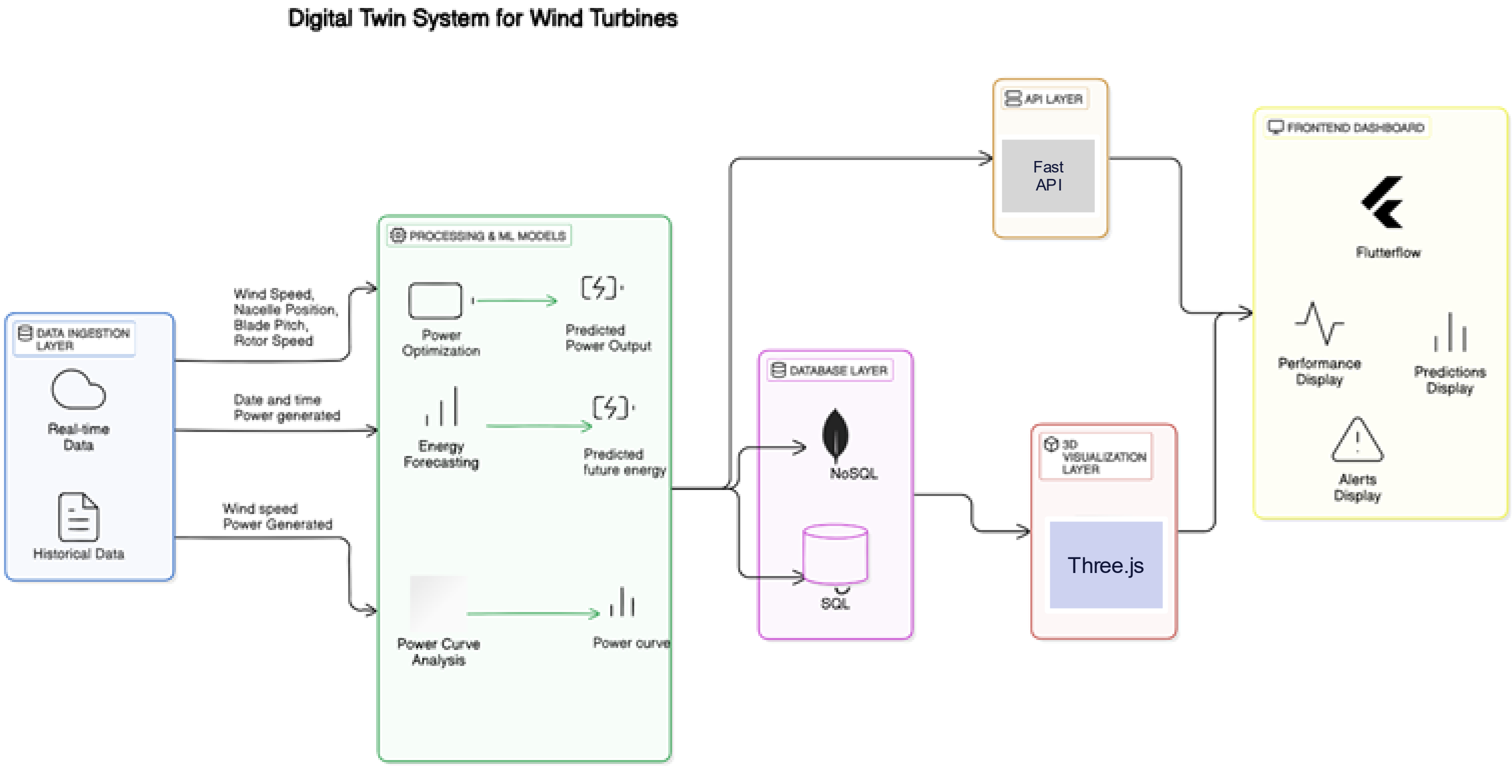
INTRODUCTION

Research Question

Optimizing Wind Turbine Performance and Forecasting Energy Output

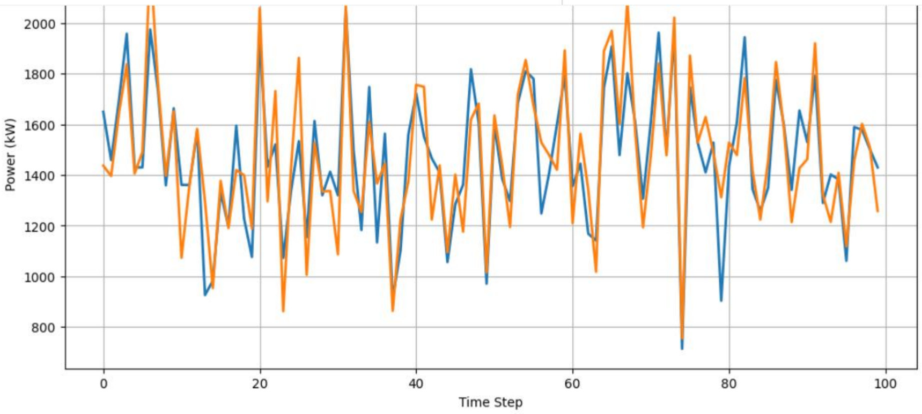
1. Gathering real-time operational data to identify performance trends and influencing factors.
2. Removeing noise, handle missing values, and extract meaningful features for modeling.
3. Using machine learning models to forecast energy output and and optimize power generation
4. Displaying live turbine metrics, predicted outputs, and alerts using intuitive dashboards.
5. Deploying models and connect to turbine systems using fast, secure APIs for real-time optimization.

System diagram



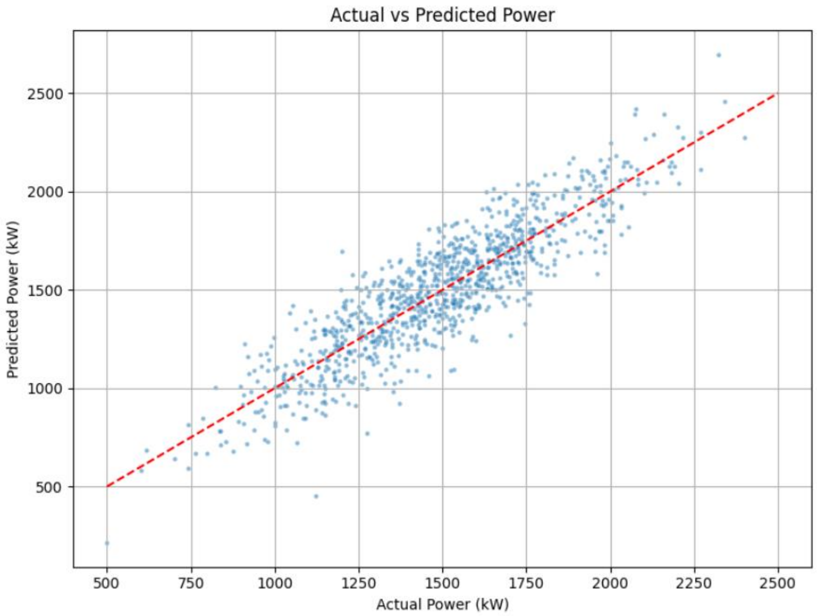
Evidence of Completion

1.Turbine Power Optimization



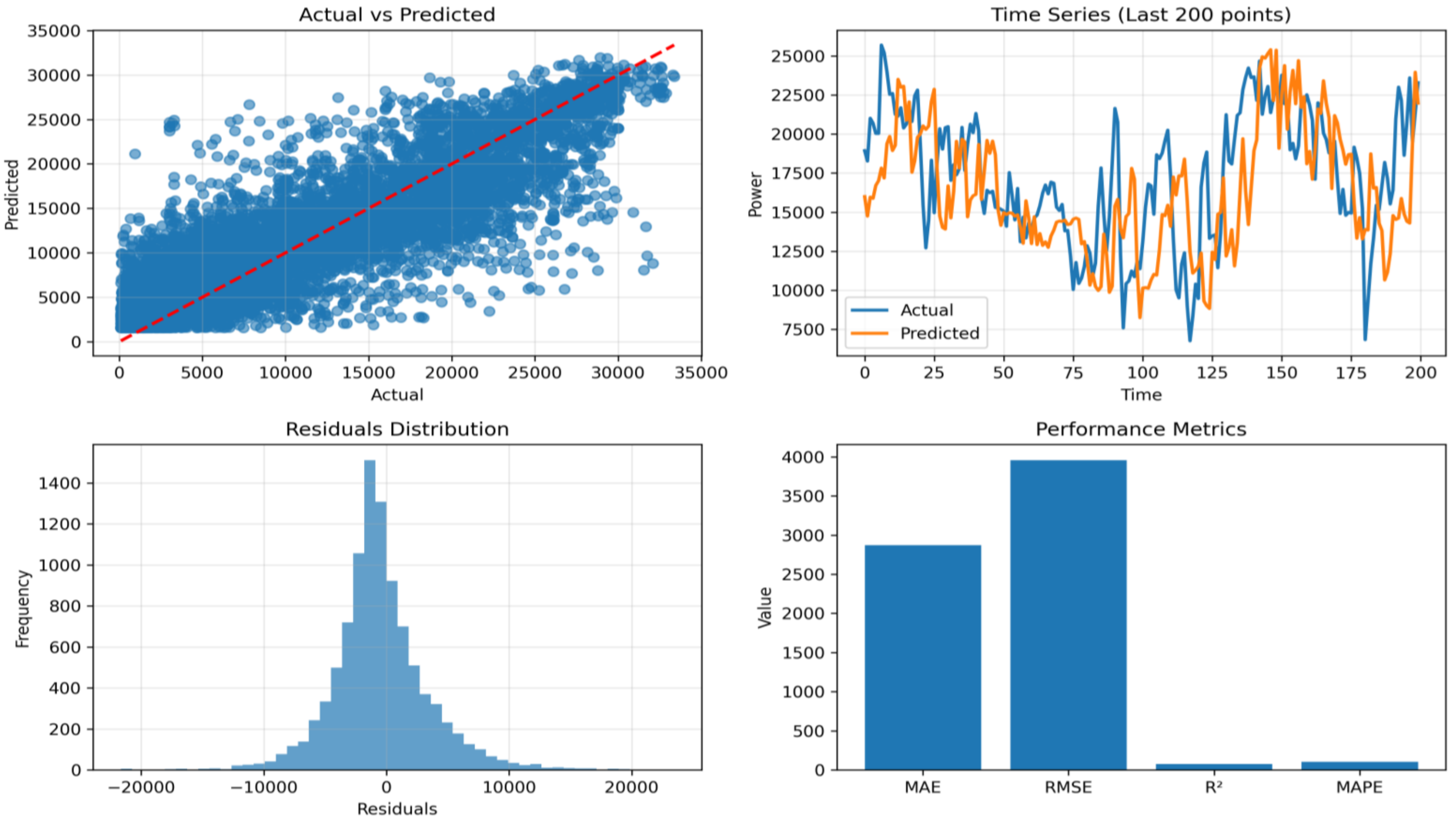
Model Performance Metrics:

	Model	R² Score	MSE	RMSE	MAE
0	XGBoost	0.91	138384	372.0	236.0
1	Gradient Boosting	0.89	149500	386.7	249.8
2	Random Forest	0.87	165210	406.4	261.5



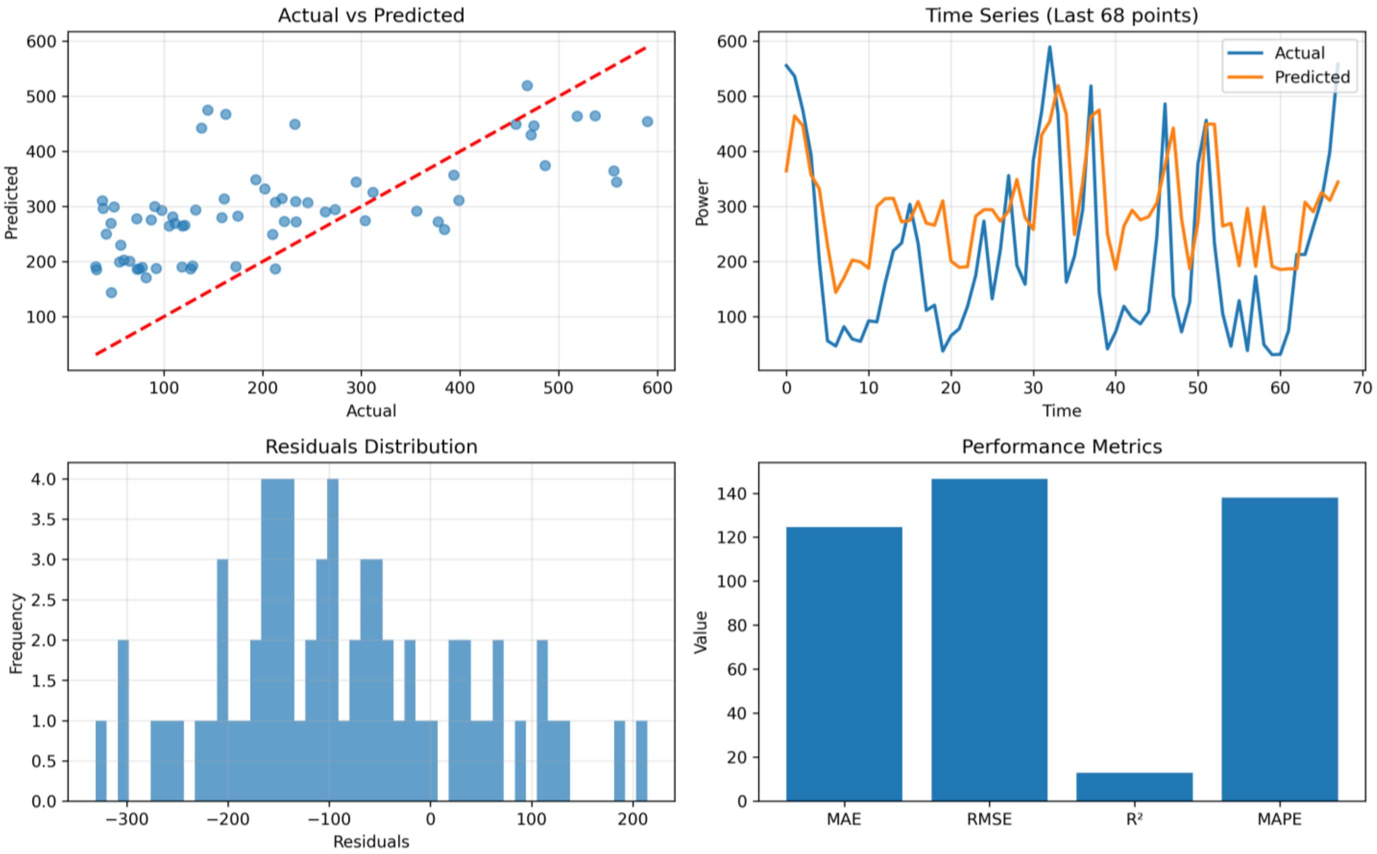
Evidence of Completion

Medium term energy forecassting (24h)

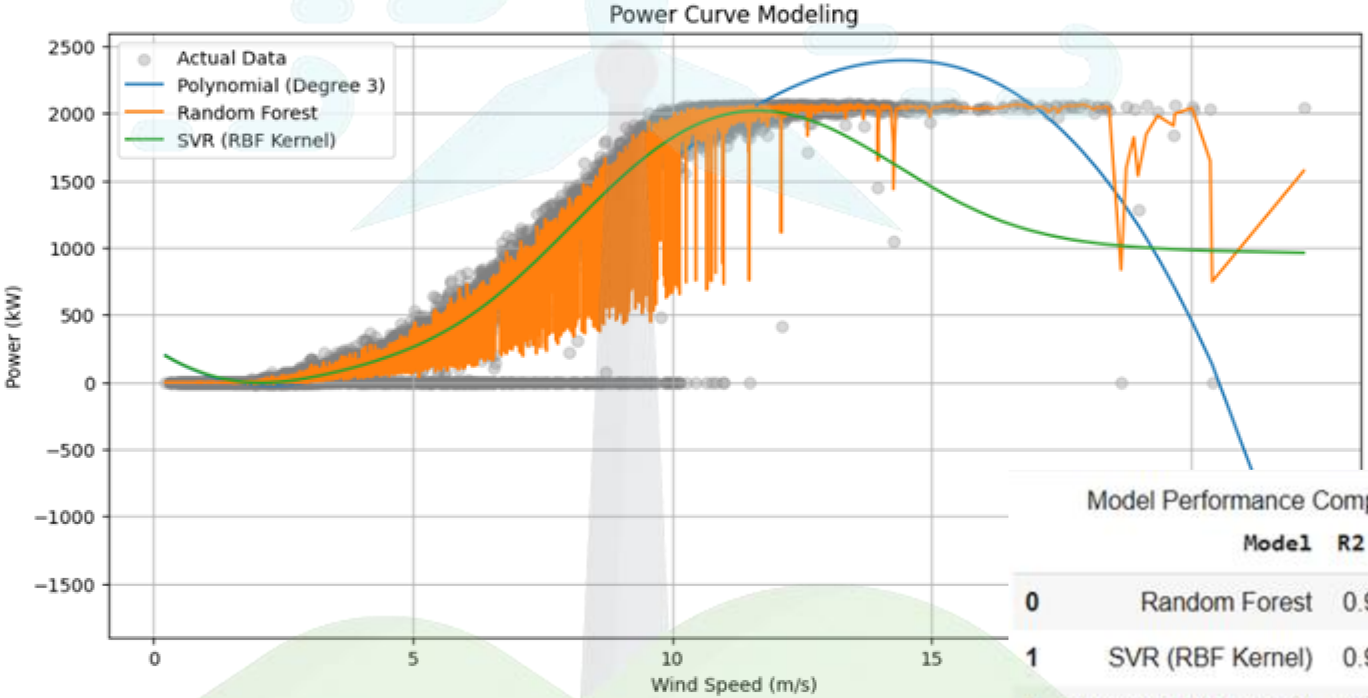


Evidence of Completion

Short term energy forecassting (6h)



Power curve analysis



Model Performance Comparison (Power Curve Analysis)

	Model	R2 Score	MSE	MAE
0	Random Forest	0.991200	3786.880000	27.460000
1	SVR (RBF Kernel)	0.957300	18290.600000	53.040000
2	Polynomial (Degree 3)	0.944800	23632.620000	80.180000

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Prediction and Optimizing Wind Turbine Noise Effect

Background

Wind turbines are a key source of renewable energy, but noise pollution from turbine operations poses environmental challenges.

This research focuses on predicting and optimizing wind turbine noise using machine learning, aiming to balance noise control with power efficiency.

Research Question

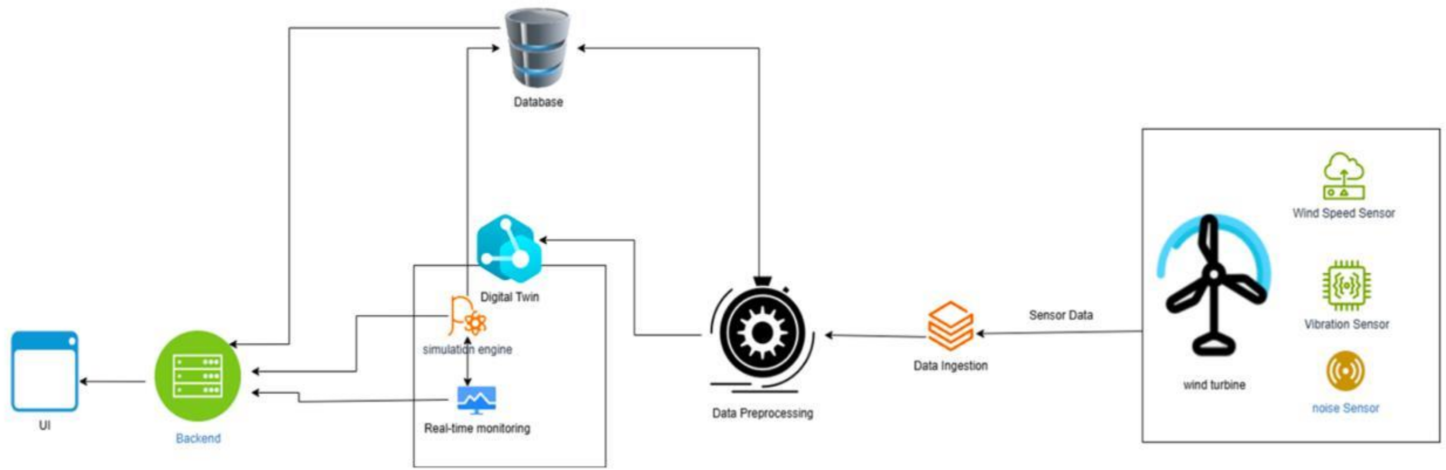
How can a digital twin system integrated with machine learning models be used to predict and optimize wind turbine noise levels while maintaining efficient power output and rotor speed under varying wind conditions?

Objectives

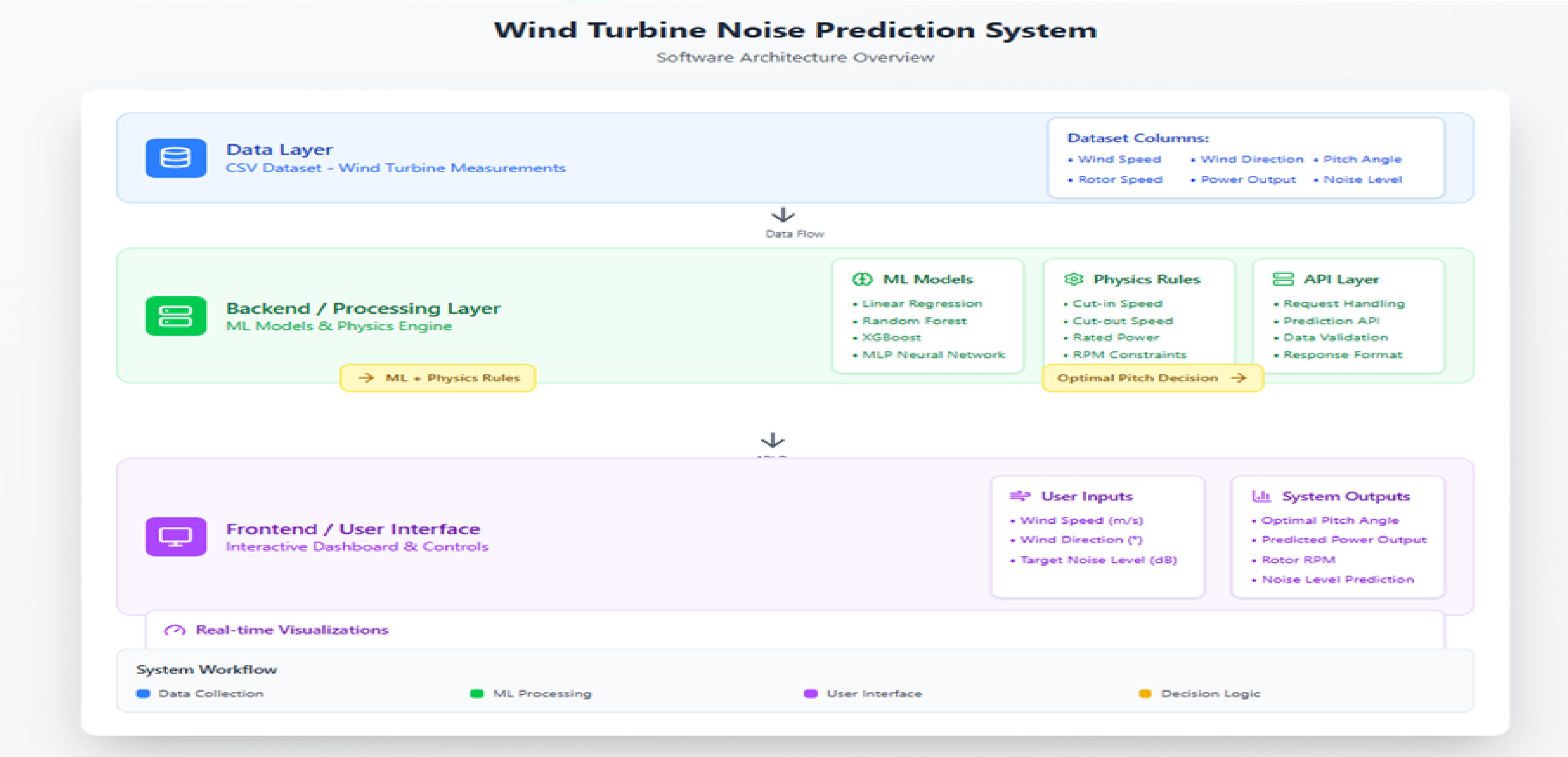
Specific Objectives:

- To identify and analyze major sources of wind turbine noise.
- To develop ML models that predict noise levels, rotor speed, power output, and pitch angle.
- To build a Digital Twin system that simulates turbine behavior based on live or manual input.
- To implement noise mitigation strategies (e.g., pitch angle adjustment)

System diagram



Methodology Diagram



Methodology

Data Collection

•Gathered sensor-based data (wind speed, wind direction, pitch angle, power output, rotor speed, noise level).

Preprocessing

•Handled missing values and outliers.

•Normalized input/output features using StandardScaler.

•Split data into training and test sets.

Methodology

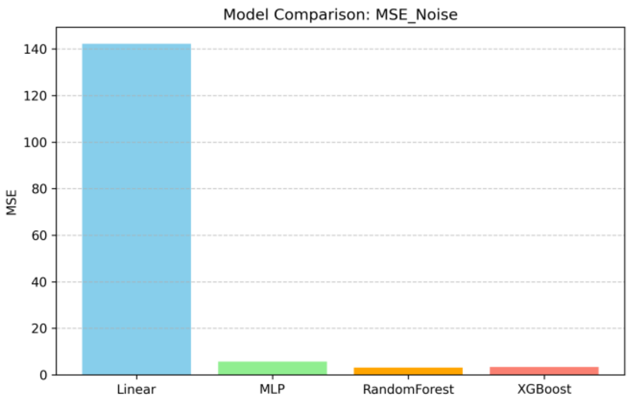
Model Training

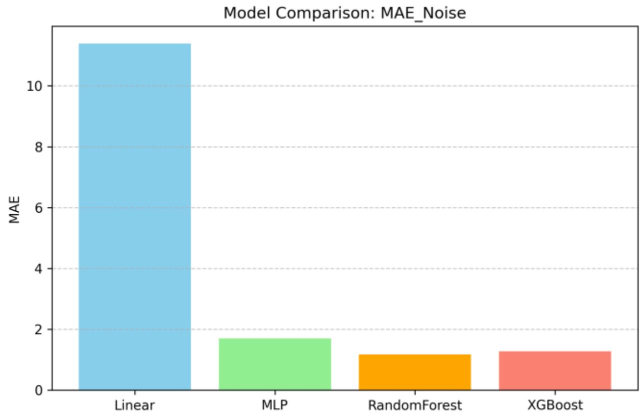
Trained multiple regression models:

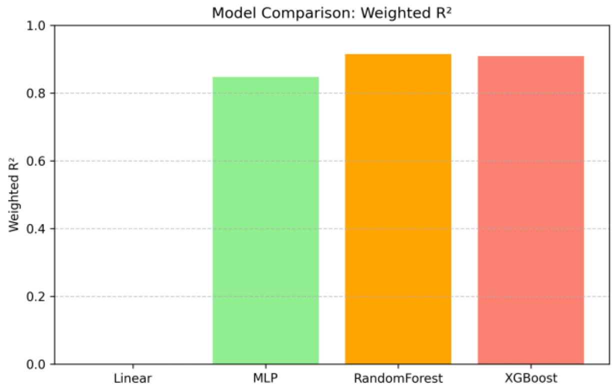
- Linear Model
- Multi-Layer Perceptron
- Random Forest (RF)
- Extreme Gradient Boosting (XGBoost)

Methodology

Model Comparison







Methodology

Model	MSE_Power	MSE_RPM	MSE_Noise	MAE_Power	MAE_RPM	MAE_Noise	R2_weighted
Linear	1750588.6	52.9	142.25	1259.92	6.94	11.39	-5.63
MLP	40248.4	1.9	5.70	131.72	0.87	1.69	0.84
RandomForest	22462.4	0.9	3.05	89.14	0.52	1.16	0.91
XGBoost	23883.8	1.06	3.35	98.33	0.59	1.27	0.90

Developed solution

3D Simulation Front-End

- Visualize turbine in 3D
- Simulates **wind speed, wind direction, RPM, and power output** based on input data

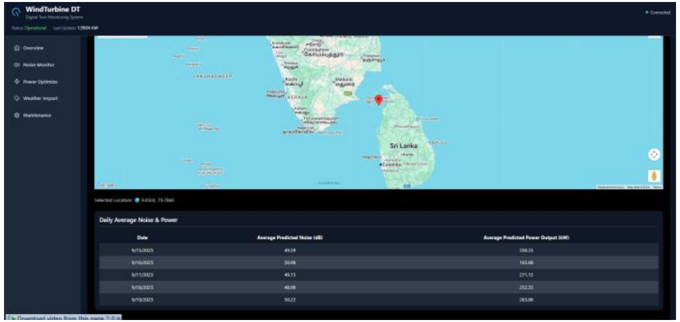
Operator Input Mode

- User enters turbine parameters
- System predicts **best pitch angle** for **maximum power output** and **minimum noise**

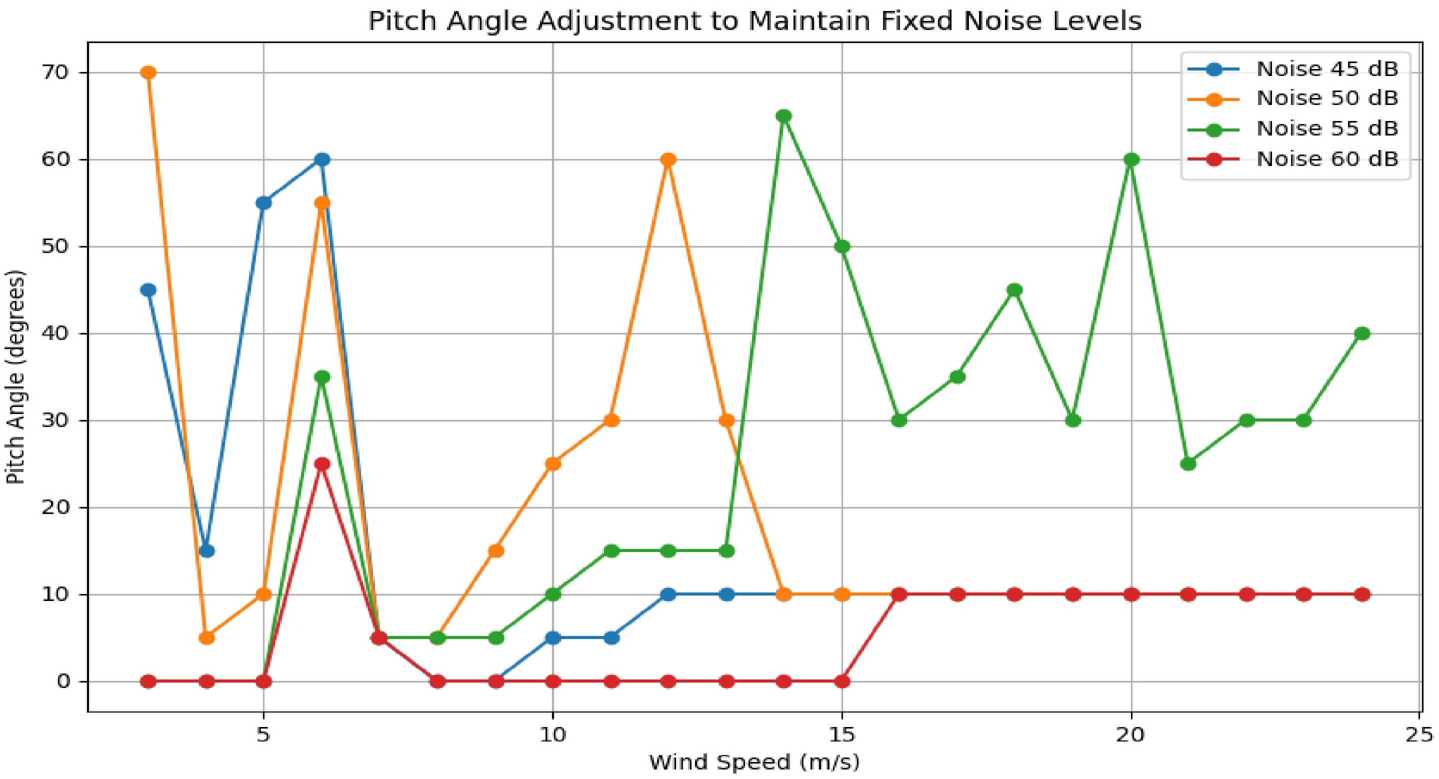
5-Day Forecast Mode

- Fetches weather data from **OpenWeather API**
- Predicts **noise levels** and **optimal power output** for next 5 days

Developed solution



Result & Discussion



Challenges Faced

Key Challenges:

Real-Time Data Availability

- Lack of access to live data from wind turbines for accurate predictions
- Had to rely on existing datasets for model training

Data Quality & Consistency

- Variations in datasets (foreign or historical) may affect model accuracy

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Dhanushikan V

Prediction and Optimizing Wind Turbine maintain Strategics

Research Question

How can digital twin technology be utilized to predict the failure, enhance the reliability, efficiency, and cost-effectiveness of wind turbine maintenance in Sri Lanka's unique tropical climate?

Specific Objective:

To design and implement a digital twin-driven maintenance strategy for wind turbines that enhances their reliability, optimizes performance, and minimizes downtime while adapting to Sri Lanka's tropical climate.

Sub Objectives:

- Collect and integrate real-time sensor data into a centralized system for continuous turbine performance monitoring.
- Use digital twin simulations to model turbine behavior under different weather conditions and analyze impacts.
- Develop machine learning models to predict failures and identify early signs of wear or degradation.
- Create weather-specific protocols like speed adjustments and optimize operations dynamically.
- Build dashboards to display turbine health and generate maintenance schedule.

Background and Literature

- Wind turbines in **tropical climates** (Sri Lanka) face issues: Nature's effect, Wear & Tear.
- Traditional maintenance →
  - Reactive: fix after breakdown → high downtime, high cost.
  - Preventive: fixed schedule → unnecessary servicing.
- **Digital Twin + Machine Learning** → condition-based and predictive maintenance.
- Literature shows:
  - LSTM useful for time-series forecasting of failures.
  - Random Forest good for fault classification.
  - Predictive maintenance reduces **downtime by ~40%** and costs by **20–25%** globally.
- Research gap: Limited **Sri Lanka-specific models** (local climate, limited datasets).

Research Problem and Objectives

Problem:

- Current maintenance in Sri Lankan wind farms is reactive/time-based → leads to failures, downtime, and high costs.

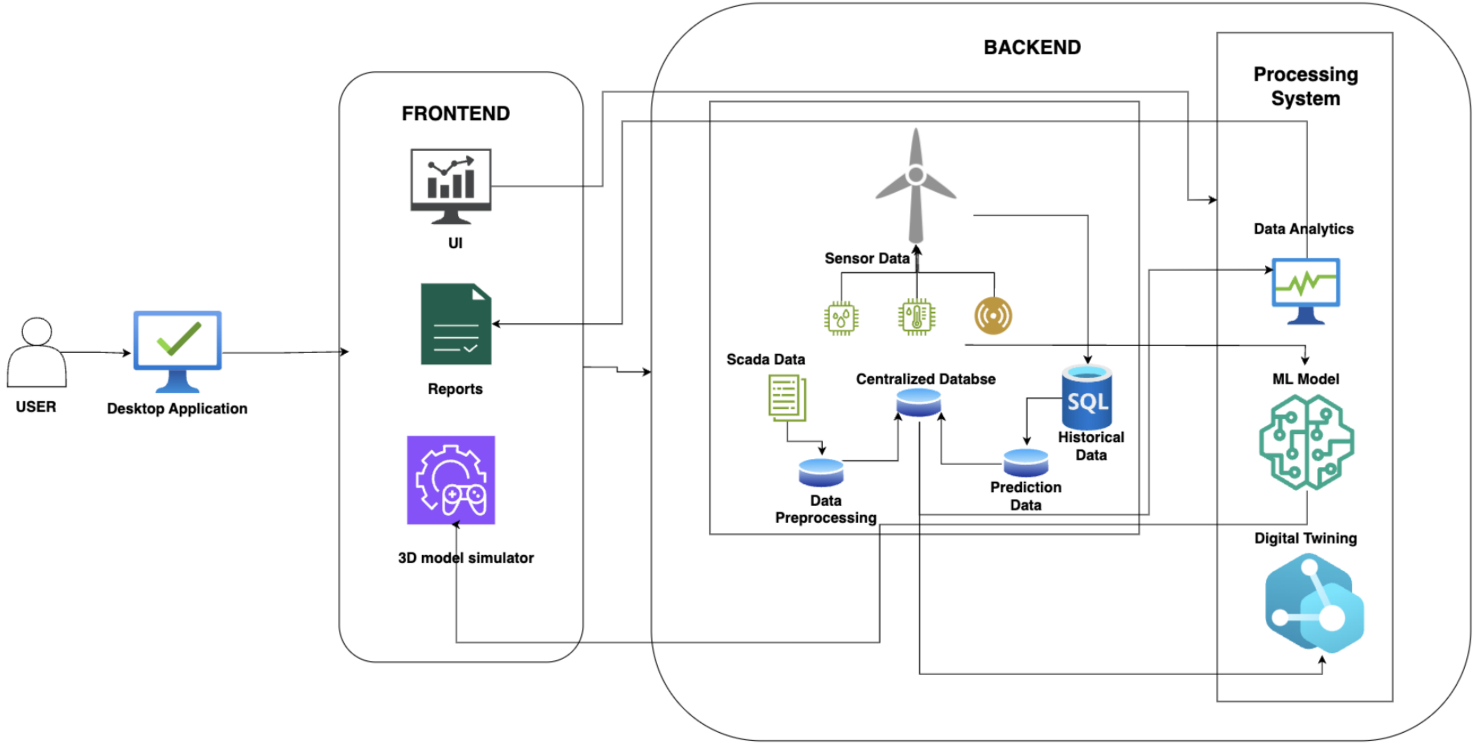
Main Objective:

- Develop a **digital twin–enabled predictive maintenance system** for tropical wind turbines.

Specific Objectives:

- Collect & integrate real-time SCADA + sensor data.
- Train ML models (LSTM, Random Forest, Ensemble).
- Predict failures + Remaining Useful Life (RUL).
- Build dynamic **health scoring system**.
- Automate maintenance scheduling & technician assignments.
- Evaluate improvements in downtime, cost, lifespan.

System Diagram and Possible Solution



Overview

3-layer architecture:

- **Frontend (React + Dashboard):** visualizes health scores, alerts, schedules.
- **Backend (FastAPI + ML Models):** runs predictions, health scoring, scheduling.
- **Data Layer (SCADA + DB):** collects, stores, manages sensor + turbine data.

Digital Twin Integration

- Create a hybrid digital twin that accepts **manual data** and **real-time inputs**.
- Provides real-time feedback required parameters.

Methodology

Data Collection:
685,000+ records, 273 features (wind speed, vibration, gearbox temp, generator temp, etc.).

Preprocessing:
Missing value handling, normalization, rolling features, feature engineering.

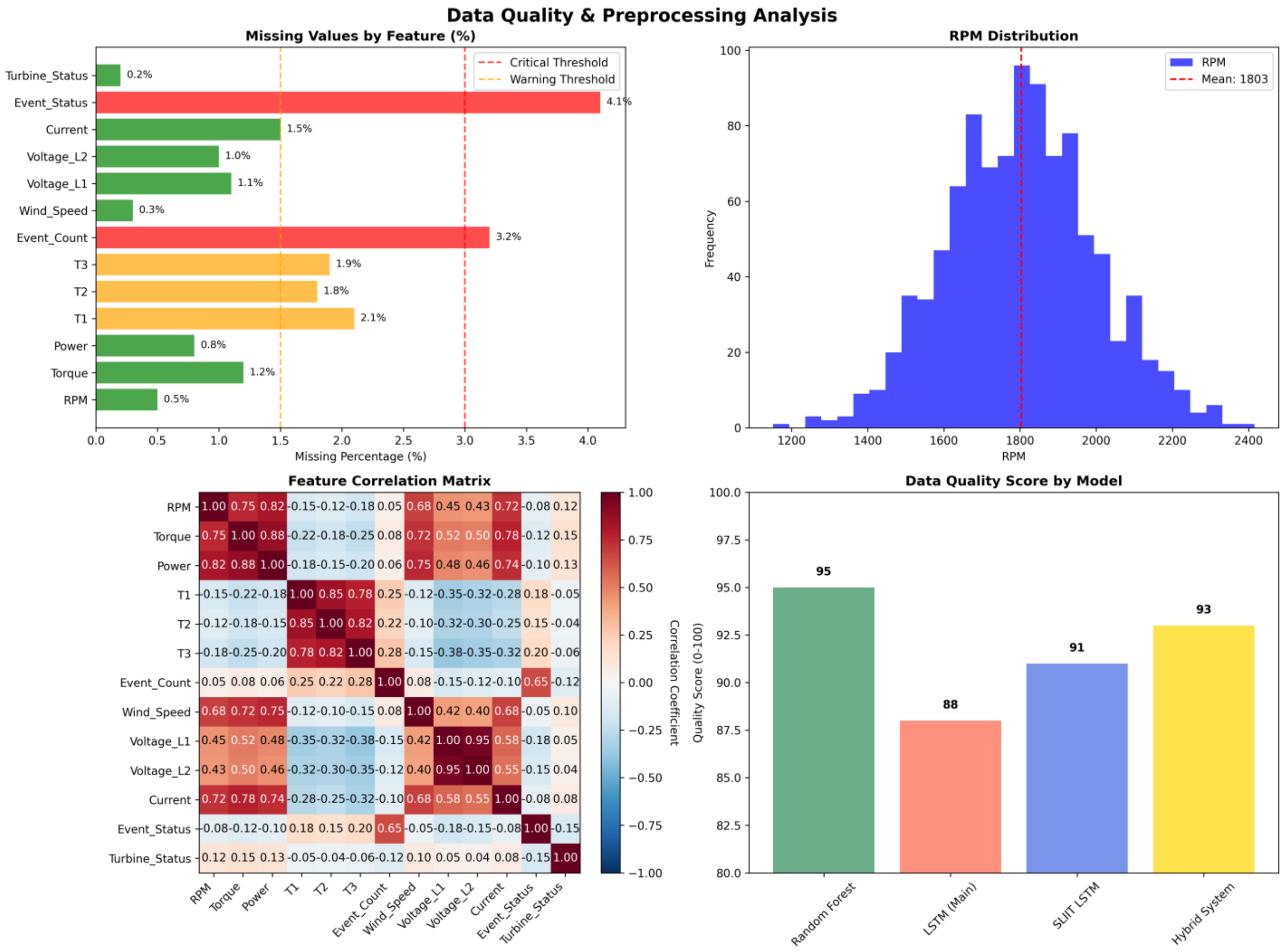
Models Tested:

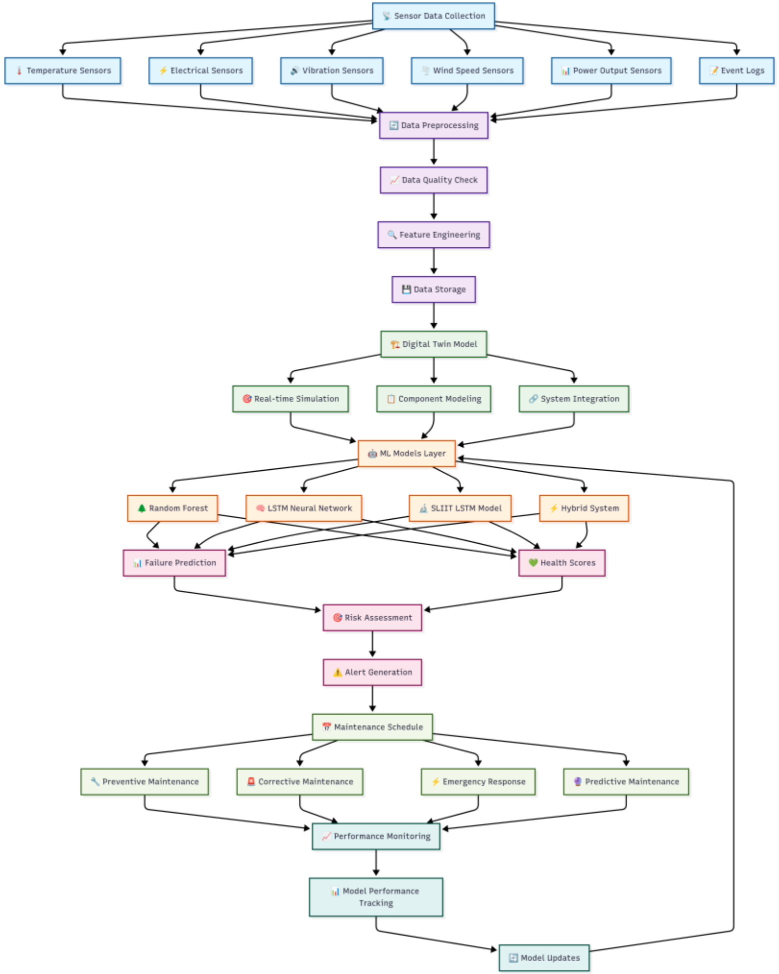
- Random Forest (fault classification).
- LSTM (time-series forecasting).
- Hybrid Ensemble (combined both → best performance).

Deployment:

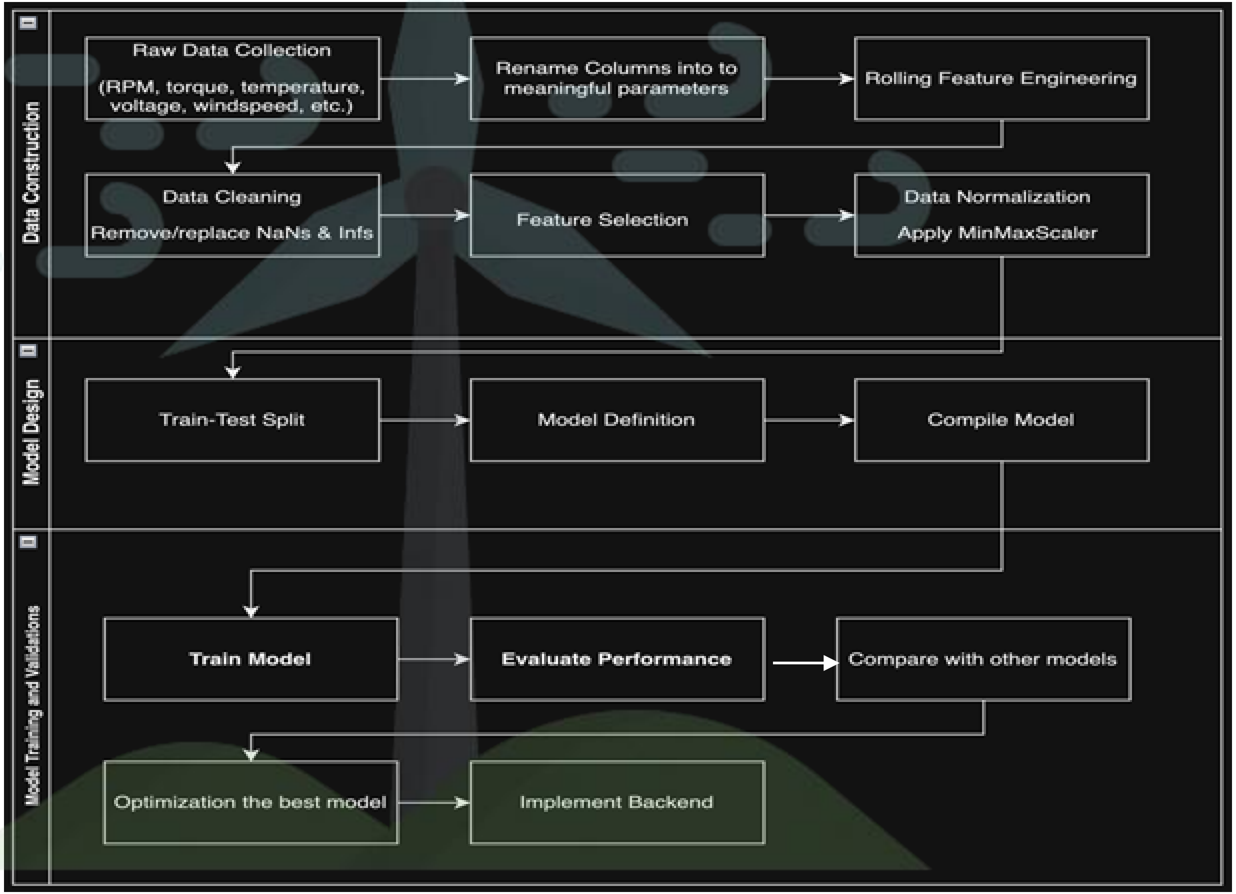
- FastAPI backend with APIs: `/predict`, `/health-scores`, `/maintenance-schedule`.
- SQLite DB for storing results.
- React Dashboard for operators.

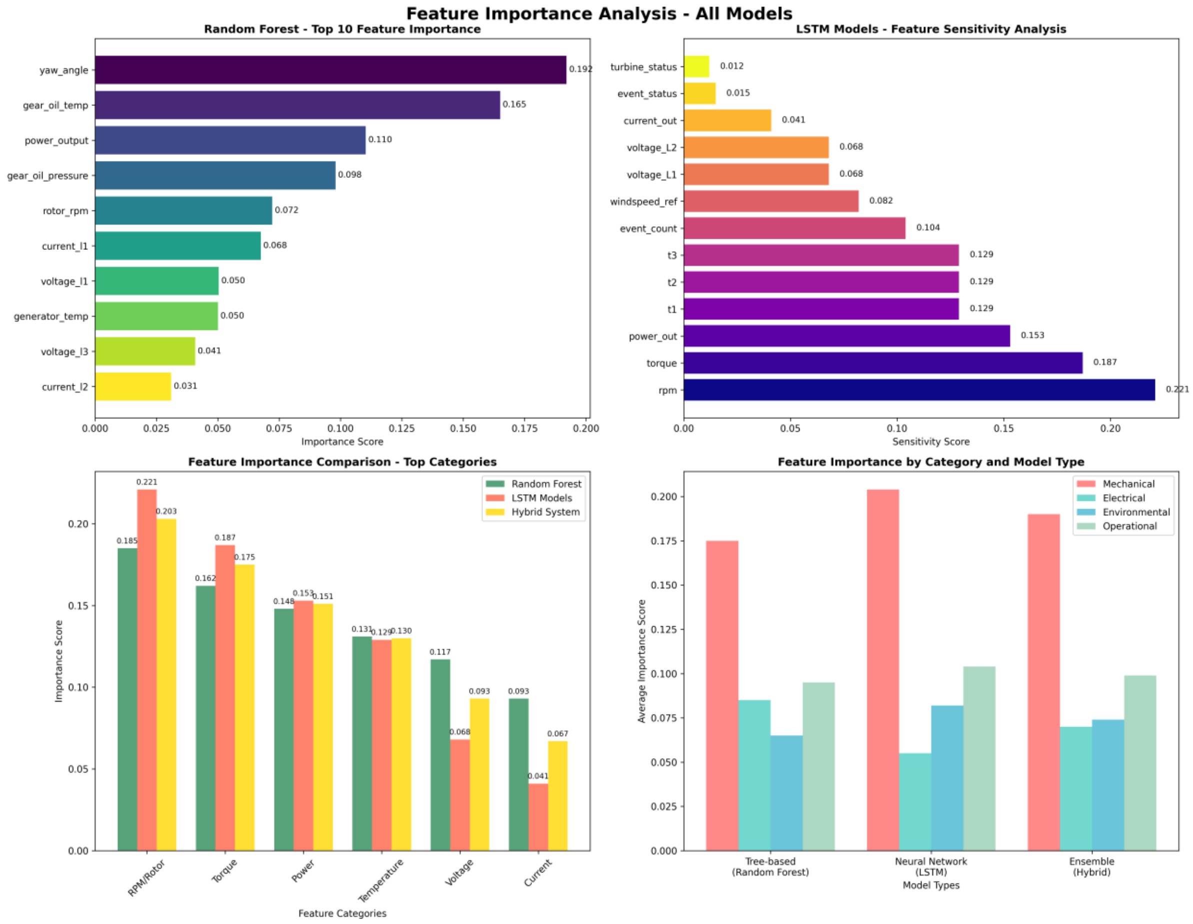
Testing:
Unit, integration, performance (<2s response, 99.5% uptime).


Data Quality & Preprocessing Analysis
Missing Values by Feature (%)
Critical Threshold
Warning Threshold
Turbine_Status 0.2%
Event_Status 4.1%
Current 1.5%
Voltage_L2 1.0%
Voltage_L1 1.1%
Wind_Speed 0.3%
Event_Count 3.2%
T3 1.9%
T2 1.8%
T1 2.1%
Power 0.8%
Torque 1.2%
RPM 0.5%
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Missing Percentage (%)
RPM Distribution
RPM
Mean: 1803
Frequency
0 20 40 60 80 100
1200 1400 1600 1800 2000 2200 2400
RPM
Feature Correlation Matrix
RPM 1.00 0.75 0.82 -0.15 -0.12 -0.18 0.05 0.68 0.45 0.43 0.72 -0.08 0.12
Torque 0.75 1.00 0.88 -0.22 -0.18 -0.25 0.08 0.72 0.52 0.50 0.78 -0.12 0.15
Power 0.82 0.88 1.00 -0.18 -0.15 -0.20 0.06 0.75 0.48 0.46 0.74 -0.10 0.13
T1 -0.15 -0.22 -0.18 1.00 0.85 0.78 0.25 -0.12 -0.35 -0.32 -0.28 0.18 -0.05
T2 -0.12 -0.18 -0.15 0.85 1.00 0.82 0.22 -0.10 -0.32 -0.30 -0.25 0.15 -0.04
T3 -0.18 -0.25 -0.20 0.78 0.82 1.00 0.28 -0.15 -0.38 -0.35 -0.32 0.20 -0.06
Event_Count 0.05 0.08 0.06 0.25 0.22 0.28 1.00 0.08 -0.15 -0.12 -0.10 0.65 -0.12
Wind_Speed 0.68 0.72 0.75 -0.12 -0.10 -0.15 0.08 1.00 0.42 0.40 0.68 -0.05 0.10
Voltage_L1 0.45 0.52 0.48 -0.35 -0.32 -0.38 -0.15 0.42 1.00 0.95 0.58 -0.18 0.05
Voltage_L2 0.43 0.50 0.46 -0.32 -0.30 -0.35 -0.12 0.40 0.95 1.00 0.55 -0.15 0.04
Current 0.72 0.78 0.74 -0.28 -0.25 -0.32 -0.10 0.68 0.58 0.55 1.00 -0.08 0.08
Event_Status -0.08 -0.12 -0.10 0.18 0.15 0.20 0.65 -0.05 -0.18 -0.15 -0.08 1.00 -0.15
Turbine_Status 0.12 0.15 0.13 -0.05 -0.04 -0.06 -0.12 0.10 0.05 0.04 0.08 -0.15 1.00
RPM Torque Power T1 T2 T3 Event_Count Wind_Speed Voltage_L1 Voltage_L2 Current Event_Status Turbine_Status
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00
Correlation Coefficient
Data Quality Score by Model
Quality Score (0-100)
80.0 82.5 85.0 87.5 90.0 92.5 95.0 97.5 100.0
95
88
91
93
Random Forest
LSTM (Main)
SLIIT LSTM
Hybrid System


Sensor Data Collection
Temperature Sensors
Electrical Sensors
Vibration Sensors
Wind Speed Sensors
Power Output Sensors
Event Logs
Data Preprocessing
Data Quality Check
Feature Engineering
Data Storage
Digital Twin Model
Real-time Simulation
Component Modeling
System Integration
ML Models Layer
Random Forest
LSTM Neural Network
SLIIT LSTM Model
Hybrid System
Failure Prediction
Health Scores
Risk Assessment
Alert Generation
Maintenance Schedule
Preventive Maintenance
Corrective Maintenance
Emergency Response
Predictive Maintenance
Performance Monitoring
Model Performance Tracking
Model Updates

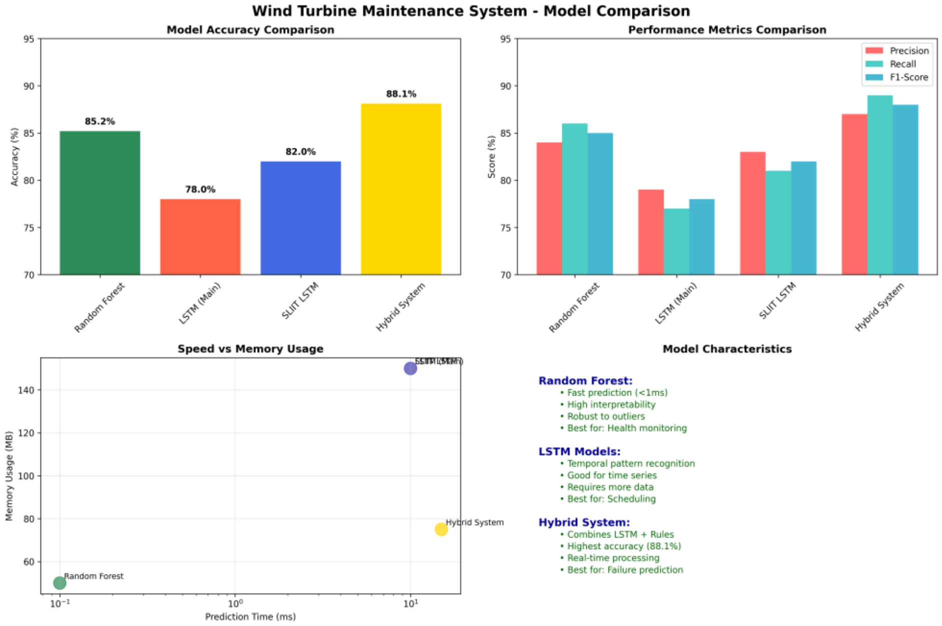
Research Methodology

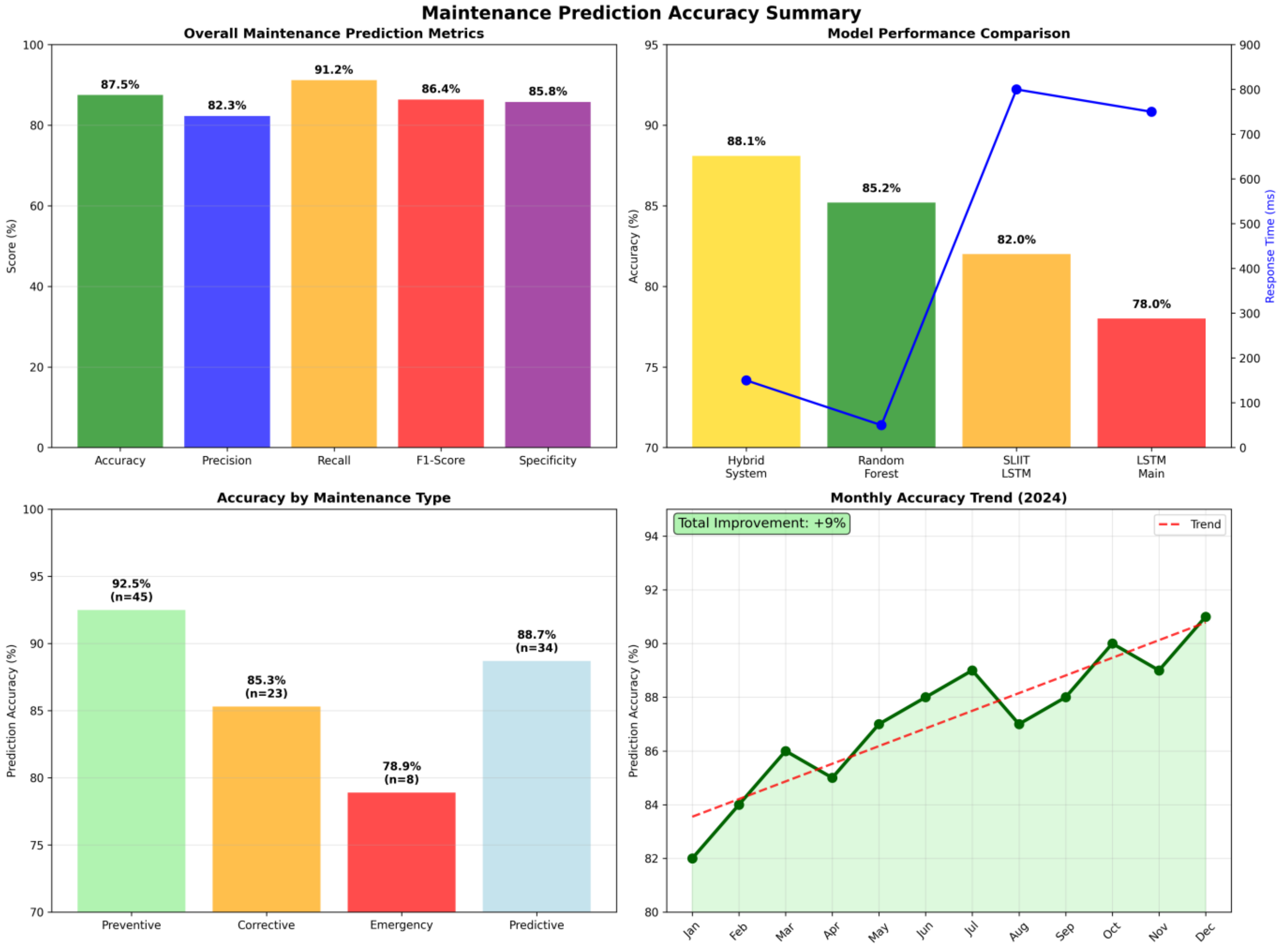



Feature Importance Analysis - All Models
Random Forest - Top 10 Feature Importance
yaw_angle 0.192
gear_oil_temp 0.165
power_output 0.110
gear_oil_pressure 0.098
rotor_rpm 0.072
current_l1 0.068
voltage_l1 0.050
generator_temp 0.050
voltage_l3 0.041
current_l2 0.031
Importance Score
LSTM Models - Feature Sensitivity Analysis
turbine_status 0.012
event_status 0.015
current_out 0.041
voltage_L2 0.068
voltage_L1 0.068
windspeed_ref 0.082
event_count 0.104
t3 0.129
t2 0.129
t1 0.129
power_out 0.153
torque 0.187
rpm 0.221
Sensitivity Score
Feature Importance Comparison - Top Categories
Random Forest
LSTM Models
Hybrid System
Importance Score
RPM/Rotor 0.185 0.221 0.203
Torque 0.162 0.187 0.175
Power 0.148 0.153 0.151
Temperature 0.131 0.129 0.130
Voltage 0.117 0.068 0.093
Current 0.093 0.041 0.067
Feature Categories
Feature Importance by Category and Model Type
Mechanical
Electrical
Environmental
Operational
Average Importance Score
Tree-based (Random Forest)
Neural Network (LSTM)
Ensemble (Hybrid)
Model Types

Developed Solution

- Predictive Maintenance API suite (FastAPI).
- Health Scoring Dashboard (shows component health %).
- Automated Scheduling (RUL-based).
- Email Notification System for technicians.
- Hybrid ML approach (LSTM + Random Forest → ensemble).





Maintenance Prediction Accuracy Summary
Overall Maintenance Prediction Metrics
Score (%)
87.5%
82.3%
91.2%
86.4%
85.8%
Accuracy
Precision
Recall
F1-Score
Specificity
Model Performance Comparison
Accuracy (%)
Response Time (ms)
88.1%
85.2%
82.0%
78.0%
Hybrid System
Random Forest
SLIIT LSTM
LSTM Main
Accuracy by Maintenance Type
Prediction Accuracy (%)
92.5% (n=45)
85.3% (n=23)
78.9% (n=8)
88.7% (n=34)
Preventive
Corrective
Emergency
Predictive
Monthly Accuracy Trend (2024)
Total Improvement: +9%
Trend
Prediction Accuracy (%)
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

Results and Discussion

Model Performance:

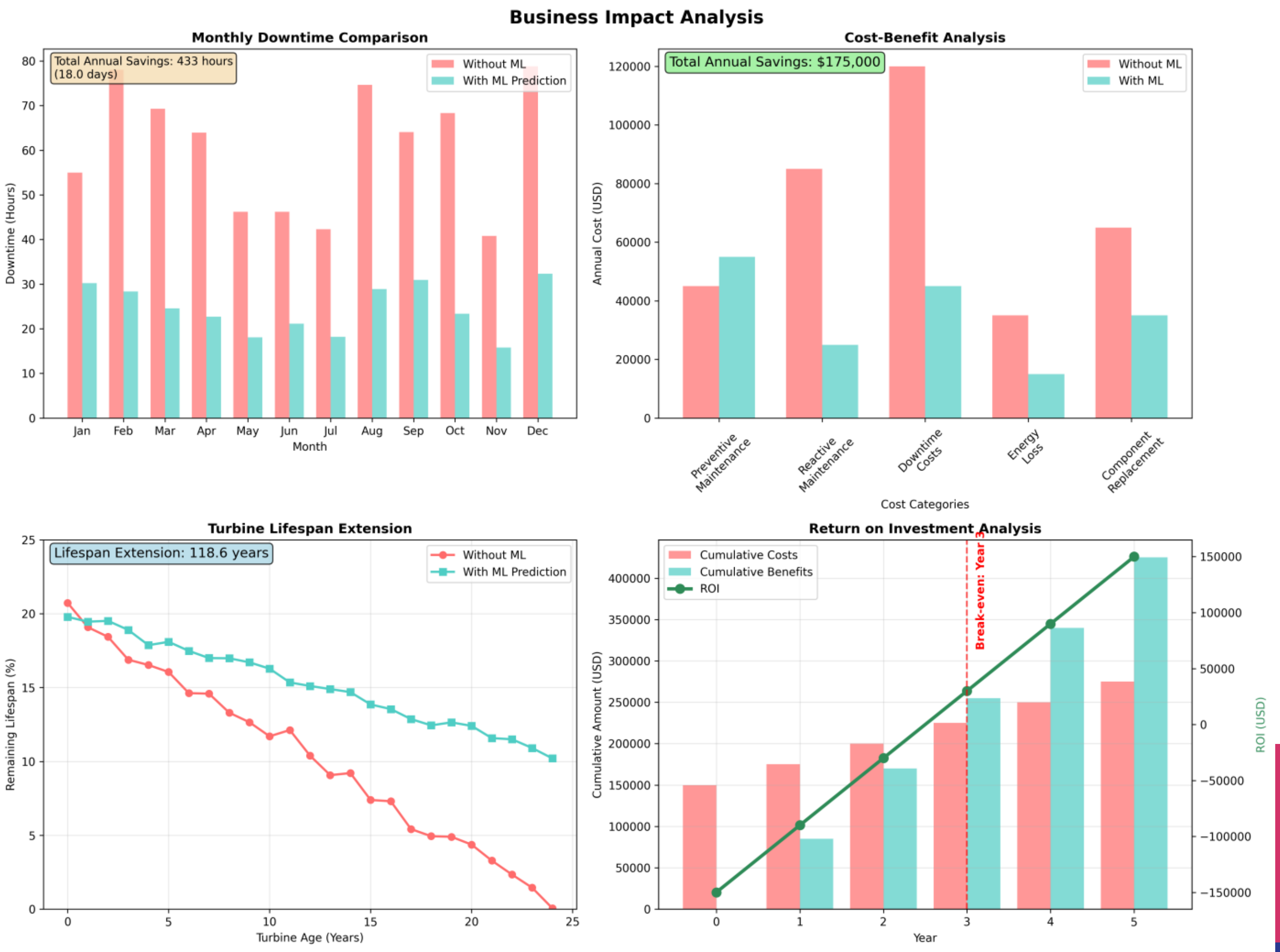
- Hybrid System Accuracy ~88% (higher than RF = 85%, LSTM = 78%).
- ROC AUC: Hybrid 0.88 > RF 0.85 > LSTM 0.78.
- Confusion Matrix shows reduced false negatives.

Prediction Results:

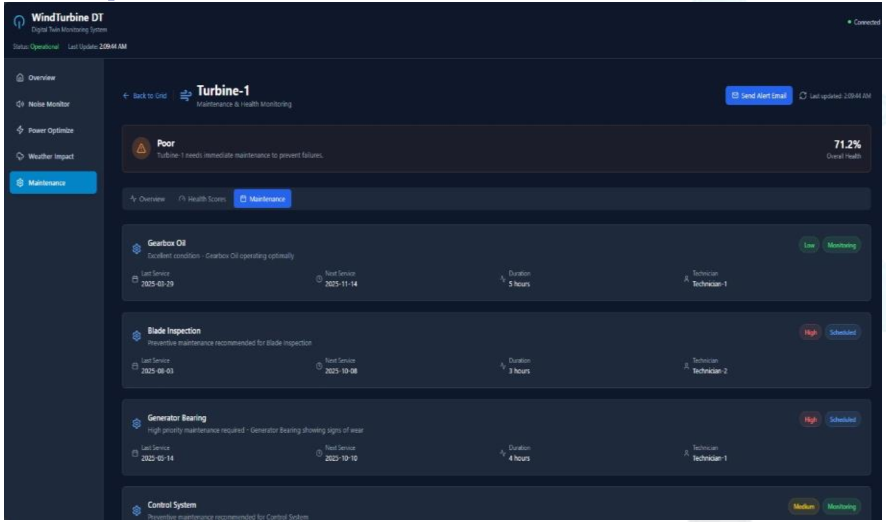
- RUL predictions align closely with actual data.
- Component health scores → Gearbox (78.5%), Generator (92.1%), etc.

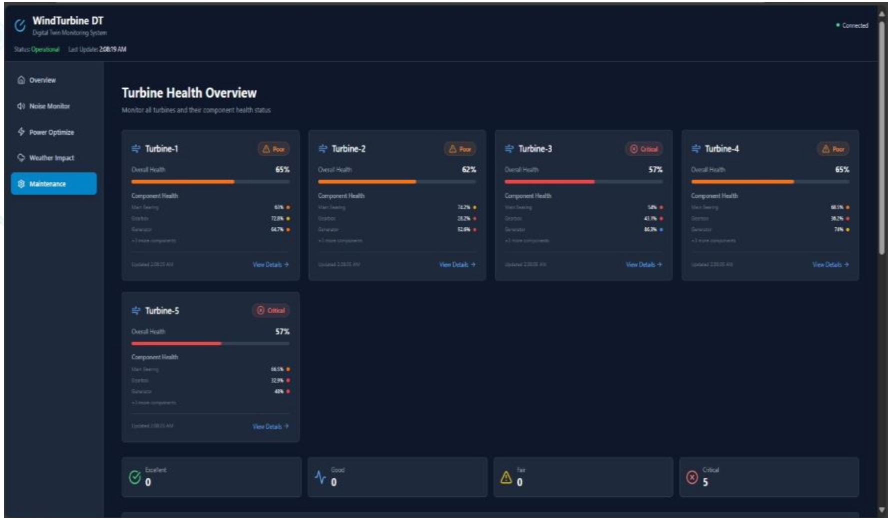
Business Impact:

- Downtime ↓ 30–40% (433 hrs saved annually).
- Costs ↓ 20–25% (~$175,000 annual savings).
- Lifespan ↑ 12–20% (gearbox, bearings, generator).
- ROI: Breakeven in ~3 years.


Business Impact Analysis
Monthly Downtime Comparison
Total Annual Savings: 433 hours (18.0 days)
Without ML
With ML Prediction
Downtime (Hours)
Month
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Cost-Benefit Analysis
Total Annual Savings: $175,000
Without ML
With ML
Annual Cost (USD)
Preventive Maintenance
Reactive Maintenance
Downtime Costs
Energy Loss
Component Replacement
Cost Categories
Turbine Lifespan Extension
Lifespan Extension: 118.6 years
Without ML
With ML Prediction
Remaining Lifespan (%)
Turbine Age (Years)
Return on Investment Analysis
Cumulative Costs
Cumulative Benefits
ROI
Break-even: Year 3
Cumulative Amount (USD)
ROI (USD)
Year

Sample UI


WindTurbine DT
Digital Twin Monitoring System
Connected
Status: Operational Last Update: 2:09:44 AM
Overview
Noise Monitor
Power Optimize
Weather Impact
Maintenance
Back to Grid
Turbine-1
Maintenance & Health Monitoring
Send Alert Email
Last updated: 2:09:44 AM
Poor
Turbine-1 needs immediate maintenance to prevent failures.
71.2%
Overall Health
Overview
Health Scores
Maintenance
Gearbox Oil
Excellent condition - Gearbox Oil operating optimally
Low
Monitoring
Last Service 2025-03-29
Next Service 2025-11-14
Duration 5 hours
Technician Technician-1
Blade Inspection
Preventive maintenance recommended for Blade Inspection
High
Scheduled
Last Service 2025-08-03
Next Service 2025-10-08
Duration 3 hours
Technician Technician-2
Generator Bearing
High priority maintenance required - Generator Bearing showing signs of wear
High
Scheduled
Last Service 2025-05-14
Next Service 2025-10-10
Duration 4 hours
Technician Technician-1
Control System
Preventive maintenance recommended for Control System
Medium
Monitoring

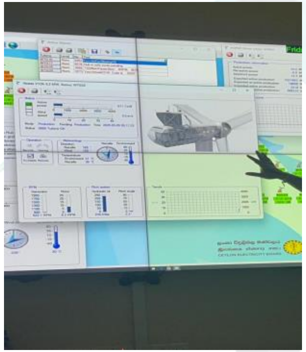

WindTurbine DT
Digital Twin Monitoring System
Connected
Status: Operational Last Update: 2:08:19 AM
Overview
Noise Monitor
Power Optimize
Weather Impact
Maintenance
Turbine Health Overview
Monitor all turbines and their component health status
Turbine-1
Poor
Overall Health 65%
Component Health
View Details
Turbine-2
Poor
Overall Health 62%
Component Health
View Details
Turbine-3
Critical
Overall Health 57%
Component Health
View Details
Turbine-4
Poor
Overall Health 65%
Component Health
View Details
Turbine-5
Critical
Overall Health 57%
Component Health
View Details
Excellent 0
Good 0
Fair 0
Critical 5

Conclusion

- Developed a **predictive maintenance system** using digital twin + ML for tropical wind turbines.
- Achieved measurable improvements:
  - Downtime ↓ 40%
  - Costs ↓ 25%
  - Lifespan ↑ 20%
- Hybrid ensemble model performed best.
- Supports **Sri Lanka's renewable energy goals (70% by 2030)**.
- Future work: collect real failure data, integrate transformers/GNN, edge-cloud hybrid deployment.







GITHUB LINK

https://github.com/it21355714/wind-turbine-optimizer.git

Thank You !